

FCC3 232

HOUSEKEEPING

Lesson 0

INTRODUCTION

0.1 QUALITY LEARNING

1. A **quality leaning process** must have **four components**:

(1) QL **imparts the knowledge** [Teaching]

(2) QL **bestows the requisite skills** [Training]

(3) QL **instills positive attitude** [Motivation]

**Necessary for
a successful
career**

2. But a **successful career** can only be realized if the student passes **the set exams**. That is the starting point.

3. Quality learning must include **coaching**
to prepares the student to pass the exams

**Fourth
component**

0.2 PARAMETERS

Quality learning can be measured using five parameters: -

1. Learning materials
2. Competent Lecturer
3. Contact time
4. Student cooperation
5. Exam performance

0.3 HOUSEKEEPING

1. The word “housekeeping” refers to “extracurricular lessons whose objective is to prepare students for the Exam”.
2. The word “Exam” as used in any house keeping lesson should be taken to mean either “CAT or the main Exam”.
3. FCC3 232 HOUSEKEEPING lessons are aimed at motivating and coaching students in preparation for the Exam.
4. The Housekeeping Lessons are divided into two parts:
Part 1 deals with motivation. Part 2 deals with coaching.

0.4 OUTLINE

FCC3 232 Housekeeping is divided into TWO PARTS

And contains TEN LESSONS as outlined below: -

- | | | |
|---------------|------------------|------------------------------|
| PART 1 | Lesson 1 | Technical Subject |
| | Lesson 2 | Time Management |
| | Lesson 3 | Things To Minimize |
| | Lesson 4 | Peer Pressure |
| | Lesson 5 | How To Study FCC3 232 |
| PART 2 | Lesson 6 | How To Pass Exam |
| | Lesson 7 | Poor Performance |
| | Lesson 8 | During The Exam |
| | Lesson 9 | Exam Design |
| | Lesson 10 | Exam Layout |

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HOUSEKEEPING

PART 1

Lesson 1

TECHNICAL SUBJECT

1.1 DEFINITION

1. A **technical subject** is defined as a subject which involves the application of specific **techniques**.
2. Usually, technical subjects have both **theoretical** and **practical** components.
3. However, some technical subjects are learned **only practically** while others are learned **theoretically**.
4. FCC3 232 is learnt **theoretically**. [The practical component is the calculations]

1.2 TECHNIQUE

1. A technique is a systematic **method** with a standard **procedure**, which is used to accomplish a specific **task**.
2. The **task** in FCC3 232 is to determine the stress induced in a structural material which is subjected to applied loads.
3. The **techniques** in FCC3 232 are the **methods** used to determine the stress. Each chapter has a specific method.
4. Every technique has **four requirements** which must be met before the learner can claim to be qualified.

1.3 REQUIREMENTS

The four requirements of a technique are: -

1. **Tools:** The notes, handouts, principles and formulae.
2. **Skill:** Skill is the ability to accomplish the **task:** -
 - (1) In the shortest time possible (**speed**)
 - (2) Using the least energy possible (**capability**)
 - (3) With the highest accuracy possible (**precision**)
3. **Attitude:** Attitude is the psychological mindset which is **necessary to acquire the requisite skill.**
4. **Practice:** “Practice makes perfect”. [**Perfect what? Ans Skill**]

1.4 MOST IMPORTANT REQUIREMENT

1. If the **tools** are given, both **attitude** and **practice** have been shown to influence **skill** in a direct way.
2. If you don't have the **correct attitude**, you cannot acquire the **required skill**.
3. If you don't **practice sufficiently**, you cannot acquire the **required skill**.
4. It can therefore be argued that **the most important requirement is skill**.

1.5 SUMMARY LESSON 1

The requirements of any technique may be summarized in the mathematical equation:

$$T_1 = T_2 S A P \dots\dots\dots \text{Eqn 1}$$

T_1 = Technique and T_2 = Tools

S = Skill

A = Attitude

P = Practice

Lesson 2

TIME MANAGEMENT

2.1 DEFINITION

1. Time management is defined as the process of **planning and control of time** in order to optimize efficiency.
2. Time management involves a deliberate effort to **rationalize the amount of time** spent on each activity.
3. The normal activities in a student's daily are: **studying, recreation, idling and socializing.**
4. The aim of time management is to ensure that the **prioritized activities are allocated sufficient time.**

2.2 WHY MANAGE TIME?

1. **Time is limited, non-elastic and non-recoverable.**
Each day must have 24 hours only. No more, No less.
2. **Time lost** doing useless things or doing nothing **can NEVER be recovered.** Time cannot be borrowed from the future.
3. **During HK Lesson 1** it was seen that the most important requirement of any technique is the **skill.**
4. **Skill is acquired after sufficient practice** which is possible only if **sufficient time** is allocated for that purpose.

2.3 HOW TO MANAGE TIME

1. The first step in time management involves **setting your priorities in the correct order.**
2. The second step in time management involves **preparing a suitable daily work program (DWP).**
3. The third step in time management involves **preparing a suitable study timetable (STT).**
4. The fourth and last step in time management involves **the deliberate effort to strictly adhere to your STT.**

2.4 SUMMARY LESSON 2

Time management can be summarized by the mathematical equation:

$$\text{TM} = \text{DWP} + \text{STT} \dots\dots\dots \text{Eqn 2}$$

TM = Time Management

DWP = Daily Work Program

STT = Study Time Table

Lesson 3

THINGS TO MINIMIZE

3.1 NORMAL ACTIVITIES

1. The normal activities in the life of a student are **studying, recreation, idling and socializing.**
2. This lesson assumes that **studying and recreation** have a higher priority than **idling and socializing.**
3. **Idling and socializing** must therefore be minimized in order to spare sufficient time for **studying and recreation.**

3.2 HOW TO MINIMIZE IDLING

1. During a normal day it is possible to have an unforeseen **free slot** in your **DWP** due to whatever reason.
2. During such a free slot, many students will just **idle** as they wait for the next activity. [“I had not planned anything”]
3. You should **identify something**, which can be done during that free slot. [For example, revise or do homework].
4. The only “**free slot**” allowed in your **DWP**, and which is very beneficial, is your allocated “**recreation time**”.

3.3 HOW TO MINIMIZE SOCIALIZING

1. **Socializing** refers to the time taken to **interact and engage with family, friends and the society.**
2. **Explain to your family, and let them understand that you are busy with your studies.**
3. **Explain to your girlfriend or boyfriend or any other friends that you are busy with your studies.**
4. **If you are in Facebook, Tweeter or Instagram consider closing or suspending your account for a while.**

3.4 SUMMARY LESSON 3

The things to be minimized can be summarized in a mathematical equation:

$$\text{TM} = \text{IS} \dots\dots\dots \text{Eqn 3}$$

T M = Things to be Minimized **I** = Idle time

S = Social affairs

Lesson 4

PEER PRESSURE

4.1 DEFINITION

1. **Peer pressure** is defined as the influence that a peer group exerts on an individual member of the peer group.
2. A peer group is a group of people who have the same or similar interests: age, background, or social status.
3. Your peers are therefore your contemporaries: age mates, college mates, classmates, neighbours and other friends.
4. Peer pressure can result in beneficial or adverse effects, depending on an individual's perception and response.

4.2 CAUSE OF PEER PRESSURE

1. Every **group** has its own **peer leaders**: individuals who are generally accepted as the “**role models**” in the group.
2. Each individual in the group will normally try to **emulate** the “**role models**”. [Nobody wants to be seen as a failure].
3. **Family, friends** and the **society** have certain **expectations** which may not be easily achievable by everyone.
4. Peer pressure arises when an individual is unable to meet the **expectations** of the **family or friends or society**.

4.3 BENEFITIAL EFFECTS

1. If taken positively, peer pressure can lead to beneficial effects on an individual's life.
2. The peer leaders form the bench marks which are used to assess the other members. [Success or Failure]
3. A positively influenced “Failure” individual will make deliberate efforts to perform better so as to catch up.
4. Many students are known to drastically improve their performance due to positive peer pressure.

4.4 ADVERSE EFFECTS

1. The most adverse effect of peer pressure is **unnecessary worry**. [**WORRY** can lead to **POST TEENAGE CRISIS**].
2. If you performed poorly in FCC3 231, **DON'T WORRY**, you have a chance to perform better in FCC3 232.
3. If you don't have pocket money, **DON'T WORRY**, you will very soon get a good job that pays well.
4. If you don't have a girlfriend/boy friend, **DON'T WORRY**, **"JUST CHILL"**, they will come pleading with you.

4.5 SUMMARY LESSON 4

The effects of peer pressure can be summarized in the mathematical equation:

$$PP \Rightarrow PTC \dots\dots\dots \text{Eqn 4}$$

PP = Peer Pressure $\Rightarrow$ means “should not lead to”

PTC = Post Teenage Crisis

Lesson 5

HOW TO STUDY

FCC3 232

5.1 SEMESTER PERIODS

The Semester can be divided into SEVEN periods

1. **Week 1: Orientation and Preparation**
2. **Week 2 to Week 7: Lectures and Study**
3. **Week 8: PoA, SLA and Revision for CATs**
4. **Week 9: CATS: Revision for each CAT**
5. **Week 10 to Week 12: Lectures and Study**
6. **Week 13: PoA, SLA and Revision for EXAMs**
7. **Weeks 14 and 15: EXAMS: Revision for each Exam**

5.2 STUDY TECHNIQUE

1. Learning Materials = Lecture Notes + Examples
2. Before each Lecture: Read the Lecture Notes
(even if you don't understand)
3. During the Lecture: Be punctual for the lecture **Have a pen and Notebook.** Listen carefully.
Take note of what is not clearly understood.

4. After the Lecture [and before end of the day]

Read the Lecture Notes again (check whether you can understand what was not clear during the lecture).

Memorize the Equations and Formulae.

Attempt and complete all examples and homework.

5. Private Study: You should have a suitable STT.

Take account of the relative “weight” of each course unit.

You should make deliberate efforts to adhere to your STT.

6. Group Study

Groups of **between 3 and 6 students** are effective.

Use your colleagues to try to understand what you could not understand during your private study.

7. Revision Technique

- 1) Copy Question to your Exercise book
- 2) Attempt the Qu without reference to the Solution
- 4) Refer to the Solution and mark your work, step by step
- 5) Take note of your mistakes and make corrections
- 6) Repeat Step 1 to Step 5 for all Questions
- 7) Repeat Step 1 to Step 6 for all Chapters

8. Exam Preparation

- 1) Attempt the Qu without reference to the Solns and record the time taken to complete the soln.
- 2) Repeat the Qu and try to increase your speed and precision.
- 3) Do sufficient practice to perfect your skill and precision to your desired level.

9. Prepare a suitable **PoA and SLA** [See Lesson 6]

10. **Avoid Forgetfulness**

5.3 FORGETFULNESS

1. Forgetfulness is a mental condition which occurs when **useful memory space** is occupied by **useless things**.
2. Useless things are like spending **too much time**:
 - (1) Chatting on **Facebook** or **Twitter** or **WhatsApp**
 - (2) Chatting with your **girlfriend** or **boyfriend**
 - (3) Watching **English** or **Spanish** football
 - (4) Dancing on **TikTok**
3. **Remember**: The memory that goes to the Exam is:
“**what you learned LESS what you forgot**”.

5.4 SUMMARY LESSON 5

The Study Technique recommended in this lesson may be summarized in two mathematical equations:

$$ST = LTA + MEF + DSP \quad \text{Eqn 5 (a)}$$

$$EM = L - F \quad \text{Eqn 5 (b)}$$

ST = Study Technique

LTA = Listen Take note and Ask

MEF = Memorize Equations and Formulae

DSP = Do Sufficient Practice

EM = Exam Memory

L = what you Learned

F = what you Forgot

THE END OF

PART 1

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PART 2

Lesson 6

HOW TO PASS EXAM

6.1 IMPORTANT INFORMATION

1. The Exam is timed.

[Speed and Precision are crucial]

2. The Exam is designed for the average student.

[Not too hard, Not too easy]

3. You are required to obtain at least 40%.

[Otherwise you are listed for a supp]

4. Supps are very RISKY.

[Avoid Supps if you can]

As you prepare
for the Exam,
keep this
information in
your mind.

6.2 PRE-REQUISITES

1. Adequate preparation
2. Suitable PoA and SLA
3. Positive attitude
4. High self confidence

**Pre-requisites
for those who
want to pass
the Exam**

6.3 PERSONAL ASSESSMENT

Assess yourself and **choose your desired target range**

1. **A1 student** = 80 – 100%
2. **Above average student** = 70 to 79%
3. **Average student** = 60 to 69%
4. **Below average student** = 50 to 59%
5. **D student** = 40 to 49%

[No student should target “below D”]

6.4 PLAN OF ACTION

Prepare a suitable **Plan of Action** as follows

1. **A1 student** = Plan to **“Attempt” 3 questions**
2. **Above average** = Plan to **“Attempt” 3 questions**
3. **Average** = Plan to **“Attempt” 2 and ½ questions**
4. **Below average** = Plan to **“Attempt” 2 questions**
5. **D student** = Plan to **“Attempt” 1 and ½ questions**

[“Attempt” means complete and get everything right]

6.5 STRATEGIC LINE OF ATTACK

Using your Plan of Action, prepare a suitable

Strategic Line of Attack as follows:

1. Choose the questions you can “**Attempt**”
2. Start with the questions you consider “**easy**”

NOTE 1. During the Exam, **Avoid CANCELLATIONS.**

NOTE 2. Items **1** and **2** are done **in advance.** [Lesson 10]

6.6 SUMMARY LESSON 6

In order to pass EXAM you must satisfy the equation:

$$\text{PE} = \text{APS} + \text{PoA} + \text{SLA} \quad \text{.....} \quad \text{Eqn 6}$$

PE = **P**ass **E**xam

APS = **A**dequate preparation

Positive attitude and **S**elf confidence

PoA = **P**lan **o**f **A**ction

SLA = **S**trategic **L**ine of **A**ttack

Lesson 7

POOR PERFORMANCE

7.1 CAUSES OF POOR PERFORMANCE

1. **Panic**
2. **Confusion**
3. **Poor Strategy**
4. **In-adequate practice**
5. **Low self confidence**

**These causes
must be
avoided at
whatever
cost**

7.2 EFFECTS OF ABOVE CAUSES

1. Time lost in **choosing questions**
2. Wrong **interpretation of questions**
3. Wrong **choice of questions**
4. Poor **speed**, Poor **precision**
5. Don't care **mentality**

Each effect
corresponds
directly to
its cause

Lesson 8

DURING THE EXAM

8.1 MUST DOs

1. **Must read the instructions carefully.**
2. **Must read each question carefully.**
3. **Must choose your questions wisely.**
4. **Must answer only what is asked.**

8.2 MUST NOTs

1. **Must NOT bring cell phones.**
2. **Must NOT bring learning materials.**
3. **Must NOT bring programmable calcs.**
4. **Must NOT bring impersonators.**
5. **Must NOT cheat in any other way.**

8.3 EXAM GUIDELINES

1. Present your work clearly.
2. Do rough work at the back pages.
3. Start each question on a fresh page.
4. Write your Registration Number clearly on the cover and at the top of each page.

Lesson 9

EXAM DESIGN

9.1 COURSE OUTLINE

1. Flexural Stress in Beams
2. Shear Stress in Beams
3. Combined Stress
4. Buckling of Struts
5. Two-Dimensional Stress
6. Strain Energy
7. Failure Theorems

The Exam is
designed to cover
the whole syllabus.

The chapters do not
have equal weight

9.2 CHAPTER COMBINATIONS

1. Chapters 1 and 2 make Question 1
2. Chapter 3 alone makes Question 2
3. Chapter 4 alone makes Question 3
4. Chapter 5 alone makes Question 4
5. Chapters 6 and 7 make Question 5

Lesson 10

EXAM LAYOUT

10.1 CAT LAYOUT

Attempt **TWO QUESTIONS ONLY**

TIME = 80 Min

QUESTION 1	Chapters 1 and 2	[15 marks]
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QUESTION 2	Chapter 3	[15 marks]
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QUESTION 3	Chapter 4	[15 marks]
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CAT FIGURES ARE PROVIDED

CAT MARKS = 30

10.2 EXAM LAYOUT

Attempt **QUESTION 1** AND **ANY OTHER TWO QUESTIONS**

TIME = 120 Min

QUESTION 1	Chapters 1 and 2	[30 marks]
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QUESTION 2	Chapter 3	[20 marks]
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QUESTION 3	Chapter 4	[20 marks]
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QUESTION 4	Chapter 5	[20 marks]
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QUESTION 5	Chapters 6 and 7	[20 marks]
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EXAM FIGURES ARE PROVIDED

EXAM MARKS = 70

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THE END